(a) Find the magnetic field and the induced electric field (in the quasistatic approximation), inside and outside the cylinder, in terms of ω, $\dot{\omega}$, and s (the distance from the axis). Calculate the torque you must exert, and from that obtain the work done per unit length ($W = \int N\, d\phi$).
(b) Use Eq. 7.44 to determine the energy stored in the resulting magnetic field.

7.3 Maxwell's Equations

7.3.1 Electrodynamics before Maxwell

So far, we have encountered the following laws, specifying the divergence and curl of electric and magnetic fields:

$$\begin{aligned} &\text{(i)} \quad \boldsymbol{\nabla} \cdot \mathbf{E} = \rho/\epsilon_0 && \text{(Gauss's law)},\\ &\text{(ii)} \quad \boldsymbol{\nabla} \cdot \mathbf{B} = 0 && \text{(no name)},\\ &\text{(iii)} \quad \boldsymbol{\nabla} \times \mathbf{E} = -\partial \mathbf{B}/\partial t && \text{(Faraday's law)},\\ &\text{(iv)} \quad \boldsymbol{\nabla} \times \mathbf{B} = \mu_0 \mathbf{J} && \text{(Ampère's law)}. \end{aligned}$$

These equations represent the state of electromagnetic theory in the mid-nineteenth century, when Maxwell began his work. They were not written in so compact a form, in those days, but their physical content was familiar. Now, it happens that there is a fatal inconsistency in these formulas. It has to do with the old rule that divergence of curl is always zero. If you apply the divergence to number (iii), everything works out:

$$\boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} \times \mathbf{E}) = \boldsymbol{\nabla} \cdot \left(-\frac{\partial \mathbf{B}}{\partial t}\right) = -\frac{\partial}{\partial t}(\boldsymbol{\nabla} \cdot \mathbf{B}).$$

The left-hand side is zero because divergence of curl is zero; the right-hand side is zero by virtue of equation (ii). But when you do the same thing to number (iv), you get into trouble:

$$\boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} \times \mathbf{B}) = \mu_0 (\boldsymbol{\nabla} \cdot \mathbf{J}); \tag{7.45}$$

the left-hand side must be zero, but the right-hand side, in general, is *not*. For *steady* currents, the divergence of $\mathbf{J}$ is zero, but when we go beyond magnetostatics Ampère's law cannot be right.

There's another way to see that Ampère's law is bound to fail for nonsteady currents. Suppose we're in the process of charging up a capacitor (Fig. 7.43). In integral form, Ampère's law reads

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}}.$$

I want to apply it to the Ampèrian loop shown in the diagram. How do I determine I_{enc}? Well, it's the total current passing through the loop, or, more precisely, the current piercing a surface that has the loop for

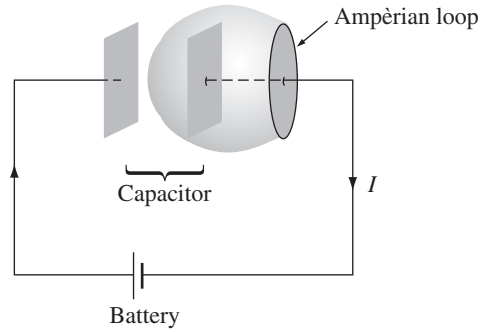


Fig. 7.43

its boundary. In this case, the *simplest* surface lies in the plane of the loop – the wire punctures this surface, so $I_{\text{enc}} = I$. Fine – but what if I draw instead the bubble-shaped surface in Fig. 7.43? *No* current passes through *this* surface, and I conclude that $I_{\text{enc}} = 0$! We never had this problem in magnetostatics because the conflict arises only when charge is piling up somewhere (in this case, on the capacitor plates). But for *nonsteady* currents (such as this one) "the current enclosed by the loop" is an ill-defined notion; it depends entirely on what surface you use. (If this seems pedantic to you – "obviously one should use the plane surface" – remember that the Ampèrian loop could be some contorted shape that doesn't even lie in a plane.)

Of course, we had no right to *expect* Ampère's law to hold outside of magnetostatics; after all, we derived it from the Biot–Savart law. However, in Maxwell's time there was no *experimental* reason to doubt that Ampère's law was of wider validity. The flaw was a purely theoretical one, and Maxwell fixed it by purely theoretical arguments.

7.3.2 How Maxwell Fixed Ampère's Law

The problem is on the right-hand side of Eq. 7.45, which *should be* zero, but *isn't*. Applying the continuity equation (Eq. 5.29) and Gauss's law, the offending term can be rewritten:

$$\boldsymbol{\nabla} \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t} = -\frac{\partial}{\partial t}(\epsilon_0 \boldsymbol{\nabla} \cdot \mathbf{E}) = -\boldsymbol{\nabla} \cdot \left(\epsilon_0 \frac{\partial \mathbf{E}}{\partial t}\right).$$

If we were to combine $\epsilon_0(\partial \mathbf{E}/\partial t)$ with $\mathbf{J}$, in Ampère's law, it would be just right to kill off the extra divergence:

$$\boxed{\boldsymbol{\nabla} \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}.} \tag{7.46}$$

(Maxwell himself had other reasons for wanting to add this quantity to Ampère's law. To him, the rescue of the continuity equation was a happy dividend rather than a primary motive. But today we recognize this

argument as a far more compelling one than Maxwell's, which was based on a now-discredited model of the ether.[24])

Such a modification changes nothing, as far as magneto*statics* is concerned: when **E** is constant, we still have $\boldsymbol{\nabla} \times \mathbf{B} = \mu_0 \mathbf{J}$. In fact, Maxwell's term is hard to detect in ordinary electromagnetic experiments, where it must compete for attention with **J** – that's why Faraday and the others never discovered it in the laboratory. However, it plays a crucial role in the propagation of electromagnetic waves, as we'll see in Chapter 9.

Apart from curing the defect in Ampère's law, Maxwell's term has a certain aesthetic appeal: just as a changing *magnetic* field induces an *electric* field (Faraday's law), so[25]

a changing electric field induces a magnetic field.

Of course, theoretical convenience and aesthetic consistency are only *suggestive* – there might, after all, be other ways to doctor up Ampère's law. The real confirmation of Maxwell's theory came in 1888 with Hertz's experiments on electromagnetic waves.

Maxwell called his extra term the **displacement current**:

$$\mathbf{J}_d \equiv \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}. \tag{7.47}$$

(It's a misleading name; $\epsilon_0(\partial \mathbf{E}/\partial t)$ has nothing to do with current, except that it adds to **J** in Ampère's law.) Let's see now how displacement current resolves the paradox of the charging capacitor (Fig. 7.43). If the capacitor plates are very close together (I didn't *draw* them that way, but the calculation is simpler if you assume this), then the electric field between them is

$$E = \frac{1}{\epsilon_0}\sigma = \frac{1}{\epsilon_0}\frac{Q}{A},$$

where Q is the charge on the plate and A is its area. Thus, between the plates,

$$\frac{\partial E}{\partial t} = \frac{1}{\epsilon_0 A}\frac{dQ}{dt} = \frac{1}{\epsilon_0 A} I.$$

Now, the "new, improved" Ampère's law (Eq. 7.46) reads, in integral form,

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}} + \mu_0 \epsilon_0 \int \left(\frac{\partial \mathbf{E}}{\partial t}\right) \cdot d\mathbf{a}. \tag{7.48}$$

If we choose the *flat* surface, then $E = 0$ and $I_{\text{enc}} = I$. If, on the other hand, we use the balloon-shaped surface, then $I_{\text{enc}} = 0$, but $\int (\partial \mathbf{E}/\partial t) \cdot d\mathbf{a} = I/\epsilon_0$. So we get the same answer for either surface, though in the first case it comes from the conduction current, and in the second from the displacement current.

[24] For the history of this subject, see A. M. Bork, *Am. J. Phys.* **31**, 854 (1963).

[25] See footnote 11 for commentary on the word "induce." The same issue arises here: Should a changing electric field be regarded as an independent source of magnetic field (along with current)? In a proximate sense it does function as a source, but since the electric field itself was produced by charges and currents, they alone are the "ultimate" sources of **E** and **B**. See S. E. Hill, *Phys. Teach.* **49**, 343 (2011); for a contrary view, see C. Savage, *Phys. Teach.* **50**, 226 (2012).

Example 7.15. Imagine two concentric metal spherical shells (Fig. 7.44). The inner one (radius a) carries a charge $Q(t)$, and the outer one (radius b) an opposite charge $-Q(t)$. The space between them is filled with ohmic material of conductivity σ, so a radial current flows:

$$\mathbf{J} = \sigma\mathbf{E} = \sigma\frac{1}{4\pi\epsilon_0}\frac{Q}{r^2}\hat{\mathbf{r}}, \quad I = -\dot{Q} = \int \mathbf{J}\cdot d\mathbf{a} = \frac{\sigma Q}{\epsilon_0}.$$

This configuration is spherically symmetrical, so the magnetic field has to be zero (the only direction it could possibly point is radial, and $\boldsymbol{\nabla}\cdot\mathbf{B} = 0 \;\Rightarrow\; \oint \mathbf{B}\cdot d\mathbf{a} = B(4\pi r^2) = 0$, so $\mathbf{B} = \mathbf{0}$). *What?* I thought currents produce magnetic fields! Isn't that what Biot–Savart and Ampère taught us? How can there be a $\mathbf{J}$ with no accompanying $\mathbf{B}$?

Solution. This is not a static configuration: Q, $\mathbf{E}$, and $\mathbf{J}$ are all functions of time; Ampère and Biot–Savart do not apply. The displacement current

$$J_d = \epsilon_0\frac{\partial\mathbf{E}}{\partial t} = \frac{1}{4\pi}\frac{\dot{Q}}{r^2}\hat{\mathbf{r}} = -\sigma\frac{Q}{4\pi\epsilon_0 r^2}\hat{\mathbf{r}}$$

exactly cancels the conduction current (in Eq. 7.46), and the magnetic field (determined by $\boldsymbol{\nabla}\cdot\mathbf{B} = 0$, $\boldsymbol{\nabla}\times\mathbf{B} = \mathbf{0}$) is indeed zero.

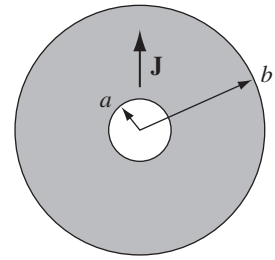


Fig. 7.44

Problem 7.35. A fat wire, radius a, carries a constant current I, uniformly distributed over its cross section. A narrow gap in the wire, of width $w \ll a$, forms a parallel-plate capacitor, as shown in Fig. 7.45. Find the magnetic field in the gap, at a distance $s < a$ from the axis.

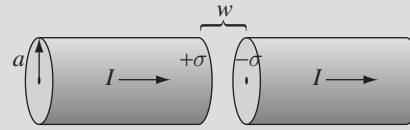


Fig. 7.45

Problem 7.36. The preceding problem was an artificial model for the charging capacitor, designed to avoid complications associated with the current spreading out over the surface of

the plates. For a more realistic model, imagine *thin* wires that connect to the centers of the plates (Fig. 7.46a). Again, the current I is constant, the radius of the capacitor is a, and the separation of the plates is $w \ll a$. Assume that the current flows out over the plates in such a way that the surface charge is uniform, at any given time, and is zero at $t = 0$.

(a) Find the electric field between the plates, as a function of t.
(b) Find the displacement current through a circle of radius s in the plane midway between the plates. Using this circle as your "Ampèrian loop," and the flat surface that spans it, find the magnetic field at a distance s from the axis.
(c) Repeat part (b), but this time use the cylindrical surface in Fig. 7.46b, which is open at the right-hand end and extends to the left through the plate and terminates outside the capacitor. Notice that the displacement current through this surface is zero, and there are two contributions to I_{enc}.[26]

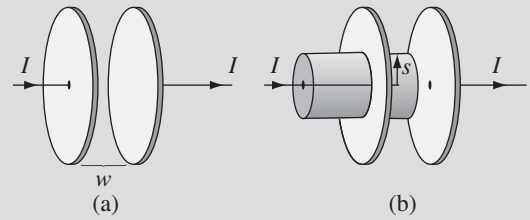


Fig. 7.46

Problem 7.37. Refer to Prob. 7.17, to which the correct answer was

$$\mathbf{E}(s,t) = \frac{\mu_0 I_0 \omega}{2\pi} \sin(\omega t) \ln\left(\frac{a}{s}\right)\hat{\mathbf{z}}.$$

(a) Find the displacement current density $\mathbf{J}_d$.
(b) Integrate it to get the total displacement current,

$$I_d = \int \mathbf{J}_d \cdot d\mathbf{a}.$$

(c) Compare I_d and I. (What's their ratio?) If the outer cylinder were, say, 2 mm in diameter, how high would the frequency have to be, for I_d to be 1 percent of I ? [This problem is designed to indicate why Faraday never discovered displacement currents, and why it is ordinarily safe to ignore them unless the frequency is extremely high.]

Problem 7.38. A conducting spherical shell (radius R, centered at the origin) is being charged by a steady current I flowing up the z-axis and connected to the "south pole," where it spreads out uniformly over the surface (see Fig. 7.47). The total charge on the sphere, at time t, is $Q(t) = It$. Find the magnetic field at a point (r, θ, ϕ) exterior to the sphere. First decide what *direction* you think $\mathbf{B}$ points; then apply the Ampère/Maxwell law in integral form

[26] This problem raises an interesting quasi-philosophical question: If you measure $\mathbf{B}$ in the laboratory, have you detected the effects of displacement current (as (b) would suggest), or merely confirmed the effects of ordinary currents (as (c) implies)? See D. F. Bartlett, *Am. J. Phys.* **58**, 1168 (1990).

(Eq. 7.48) to an appropriate Ampèrian loop. In fact, do it three different ways: using (as the surface bounded by the Ampèrian loop) (a) a spherical cap north of the loop, (b) a spherical bowl south of the loop, and (c) a cone with vertex at the origin. (d) What does your result say for θ very near to π?[27]

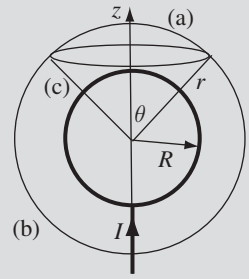


Fig. 7.47

7.3.3 Maxwell's Equations

In the last section we put the finishing touches on Maxwell's equations:[28]

$$\begin{aligned}
&\text{(i)} \quad \boldsymbol{\nabla}\cdot\mathbf{E} = \frac{1}{\epsilon_0}\rho && \text{(Gauss's law)},\\
&\text{(ii)} \quad \boldsymbol{\nabla}\cdot\mathbf{B} = 0 && \text{(no name)},\\
&\text{(iii)} \quad \boldsymbol{\nabla}\times\mathbf{E} = -\frac{\partial\mathbf{B}}{\partial t} && \text{(Faraday's law)},\\
&\text{(iv)} \quad \boldsymbol{\nabla}\times\mathbf{B} = \mu_0\mathbf{J} + \mu_0\epsilon_0\frac{\partial\mathbf{E}}{\partial t} && \text{(Ampère's law with Maxwell's correction)}.
\end{aligned} \tag{7.49}$$

Together with the Lorentz force law,

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v}\times\mathbf{B}), \tag{7.50}$$

they summarize the entire theoretical content of classical electrodynamics[29] (save for some special properties of matter, which we encountered in Chapters 4 and 6). Even the continuity equation,

$$\boldsymbol{\nabla}\cdot\mathbf{J} = -\frac{\partial\rho}{\partial t}, \tag{7.51}$$

[27] T. H. Boyer, *Am. J. Phys.* **87**, 729 (2019).

[28] Maxwell didn't have access to modern vector notation (in particular, the del operator), and in his formulation there were actually 20 equations – including the potentials as well as the fields. It was Oliver Heaviside who distilled them into the tidy package we now know as "Maxwell's equations." See B. J. Hunt, *Physics Today* November 2012, p. 48.

[29] Like any differential equations, Maxwell's must be supplemented by suitable *boundary conditions*. Because these are typically "obvious" from the context (for example, **E** and **B** go to zero at large distances from a localized charge distribution), it is easy to forget that they play an essential role.

which is the mathematical expression of conservation of charge, can be derived from Maxwell's equations by applying the divergence to number (iv).

I have written Maxwell's equations in the traditional way, which emphasizes that they specify the divergence and curl of **E** and **B**. In this form, they reinforce the notion that electric fields can be produced *either* by charges (ρ) *or* by changing magnetic fields ($\partial\mathbf{B}/\partial t$), and magnetic fields can be produced *either* by currents (**J**) *or* by changing electric fields ($\partial\mathbf{E}/\partial t$). Actually, this is misleading, because $\partial\mathbf{B}/\partial t$ and $\partial\mathbf{E}/\partial t$ are *themselves* due to charges and currents. I think it is logically preferable to write

$$\left.\begin{aligned} &\text{(i)}\ \ \boldsymbol{\nabla}\cdot\mathbf{E} = \frac{1}{\epsilon_0}\rho, \qquad \text{(iii)}\ \ \boldsymbol{\nabla}\times\mathbf{E} + \frac{\partial\mathbf{B}}{\partial t} = \mathbf{0}, \\ &\text{(ii)}\ \ \boldsymbol{\nabla}\cdot\mathbf{B} = 0, \qquad \text{(iv)}\ \ \boldsymbol{\nabla}\times\mathbf{B} - \mu_0\epsilon_0\frac{\partial\mathbf{E}}{\partial t} = \mu_0\mathbf{J}, \end{aligned}\right\} \tag{7.52}$$

with the fields (**E** and **B**) on the left and the sources (ρ and **J**) on the right. This notation emphasizes that all electromagnetic fields are ultimately attributable to charges and currents. Maxwell's equations tell you how *charges* produce *fields*; reciprocally, the force law tells you how *fields* affect *charges*.

Problem 7.39. Suppose

$$\mathbf{E}(\mathbf{r},t) = \frac{1}{4\pi\epsilon_0}\frac{q}{r^2}\theta(vt-r)\hat{\mathbf{r}}, \qquad \mathbf{B}(\mathbf{r},t) = \mathbf{0}.$$

(The theta function is defined in Prob. 1.46b.) Show that these fields satisfy all of Maxwell's equations, and determine ρ and **J**. Describe the physical situation that gives rise to these fields.

7.3.4 Magnetic Charge

There is a pleasing symmetry to Maxwell's equations; it is particularly striking in free space, where ρ and **J** vanish:

$$\left.\begin{aligned} &\boldsymbol{\nabla}\cdot\mathbf{E} = 0, \quad \boldsymbol{\nabla}\times\mathbf{E} = -\frac{\partial\mathbf{B}}{\partial t}, \\ &\boldsymbol{\nabla}\cdot\mathbf{B} = 0, \quad \boldsymbol{\nabla}\times\mathbf{B} = \mu_0\epsilon_0\frac{\partial\mathbf{E}}{\partial t}. \end{aligned}\right\}$$

If you replace **E** by **B** and **B** by $-\mu_0\epsilon_0\mathbf{E}$, the first pair of equations turns into the second, and vice versa. This symmetry[30] between **E** and **B** is spoiled, though, by the charge term in Gauss's law and the current term in Ampère's law. You can't help wondering why the corresponding quantities are "missing" from $\boldsymbol{\nabla}\cdot\mathbf{B} = 0$ and $\boldsymbol{\nabla}\times\mathbf{E} = -\partial\mathbf{B}/\partial t$. What if we had

[30] Don't be distracted by the pesky constants μ_0 and ϵ_0; these are present only because the SI system measures **E** and **B** in different units, and would not occur, for instance, in the Gaussian system.

$$\left.\begin{aligned} &\text{(i)}\ \ \boldsymbol{\nabla}\cdot\mathbf{E}=\frac{1}{\epsilon_0}\rho_e, \qquad \text{(iii)}\ \ \boldsymbol{\nabla}\times\mathbf{E}=-\mu_0\mathbf{J}_m-\frac{\partial\mathbf{B}}{\partial t},\\ &\text{(ii)}\ \ \boldsymbol{\nabla}\cdot\mathbf{B}=\mu_0\rho_m, \qquad \text{(iv)}\ \ \boldsymbol{\nabla}\times\mathbf{B}=\mu_0\mathbf{J}_e+\mu_0\epsilon_0\frac{\partial\mathbf{E}}{\partial t}. \end{aligned}\right\} \tag{7.53}$$

Then ρ_m would represent the density of magnetic "charge," and ρ_e the density of electric charge; $\mathbf{J}_m$ would be the current of magnetic charge, and $\mathbf{J}_e$ the current of electric charge. Both charges would be conserved:

$$\boldsymbol{\nabla}\cdot\mathbf{J}_m=-\frac{\partial\rho_m}{\partial t} \quad \text{and} \quad \boldsymbol{\nabla}\cdot\mathbf{J}_e=-\frac{\partial\rho_e}{\partial t}. \tag{7.54}$$

The former follows by application of the divergence to (iii), the latter by taking the divergence of (iv).

In a sense, Maxwell's equations *beg* for magnetic charge to exist – it would fit in so nicely. And yet, in spite of a diligent search, no one has ever found any.[31] As far as we know, ρ_m is zero everywhere, and so is $\mathbf{J}_m$; $\mathbf{B}$ is *not* on equal footing with $\mathbf{E}$: there exist stationary sources for $\mathbf{E}$ (electric charges) but none for $\mathbf{B}$. (This is reflected in the fact that magnetic multipole expansions have no monopole term, and magnetic dipoles consist of current loops, not separated north and south "poles.") Apparently God just didn't *make* any magnetic charge. (In *quantum* electrodynamics, by the way, it's a more than merely aesthetic shame that magnetic charge does not seem to exist: Dirac showed that the existence of *magnetic* charge would explain why *electric* charge is *quantized*. See Prob. 8.21.)

Problem 7.40. Assuming that "Coulomb's law" for magnetic charges (q_m) reads

$$\mathbf{F}=\frac{\mu_0}{4\pi}\frac{q_{m_1}q_{m_2}}{\mathcal{r}^2}\hat{\boldsymbol{\mathcal{r}}}, \tag{7.55}$$

work out the force law for a monopole q_m moving with velocity $\mathbf{v}$ through electric and magnetic fields $\mathbf{E}$ and $\mathbf{B}$.[32]

Problem 7.41. Suppose a magnetic monopole q_m passes through a resistanceless loop of wire with self-inductance L. What current is induced in the loop?[33]

7.3.5 Maxwell's Equations in Matter

Maxwell's equations in the form 7.49 are complete and correct as they stand. However, when you are working with materials that are subject to electric and magnetic polarization there is a more convenient way to *write*

[31] For an extensive bibliography, see A. S. Goldhaber and W. P. Trower, *Am. J. Phys.* **58**, 429 (1990).
[32] For interesting commentary, see W. Rindler, *Am. J. Phys.* **57**, 993 (1989).
[33] This is one of the methods used to search for monopoles in the laboratory; see B. Cabrera, *Phys. Rev. Lett*. **48**, 1378 (1982).

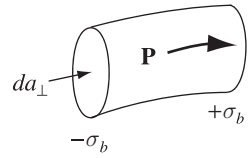


Fig. 7.48

them. For inside polarized matter there will be accumulations of "bound" charge and current, over which you exert no direct control. It would be nice to reformulate Maxwell's equations so as to make explicit reference only to the "free" charges and currents.

We have already learned, from the static case, that an electric polarization **P** produces a bound charge density

$$\rho_b = -\boldsymbol{\nabla} \cdot \mathbf{P} \tag{7.56}$$

(Eq. 4.12). Likewise, a magnetic polarization (or "magnetization") **M** results in a bound current

$$\mathbf{J}_b = \boldsymbol{\nabla} \times \mathbf{M} \tag{7.57}$$

(Eq. 6.13). There's just one new feature to consider in the *non*static case: any *change* in the electric polarization involves a *flow* of (bound) charge, which must be included in the total current. For suppose we examine a tiny chunk of polarized material (Fig. 7.48). The polarization introduces a charge density $\sigma_b = P$ at one end and $-\sigma_b$ at the other (Eq. 4.11). If P now *increases* a bit, the charge on each end increases accordingly, giving a net current

$$dI = \frac{\partial \sigma_b}{\partial t}\, da_\perp = \frac{\partial P}{\partial t}\, da_\perp.$$

The current density, therefore, is

$$\mathbf{J}_p = \frac{\partial \mathbf{P}}{\partial t}. \tag{7.58}$$

This **polarization current** has nothing to do with the *bound* current $\mathbf{J}_b$. The latter is associated with *magnetization* of the material and involves the spin and orbital motion of electrons; $\mathbf{J}_p$, by contrast, is the result of the linear motion of charge when the electric polarization changes. If **P** points to the right, and is increasing, then each plus charge moves a bit to the right and each minus charge to the left; the cumulative effect is the polarization current $\mathbf{J}_p$.

We ought to check that Eq. 7.58 is consistent with the continuity equation:

$$\boldsymbol{\nabla} \cdot \mathbf{J}_p = \boldsymbol{\nabla} \cdot \frac{\partial \mathbf{P}}{\partial t} = \frac{\partial}{\partial t}(\boldsymbol{\nabla} \cdot \mathbf{P}) = -\frac{\partial \rho_b}{\partial t}.$$

Yes: The continuity equation *is* satisfied; in fact, $\mathbf{J}_p$ is essential to ensure the conservation of bound charge. (Incidentally, a changing *magnetization* does *not* lead to any analogous accumulation of charge or current. The bound

current $\mathbf{J}_b = \boldsymbol{\nabla} \times \mathbf{M}$ varies in response to changes in $\mathbf{M}$, to be sure, but that's about it.)

In view of all this, the total charge density can be separated into two parts:

$$\rho = \rho_f + \rho_b = \rho_f - \boldsymbol{\nabla} \cdot \mathbf{P}, \tag{7.59}$$

and the current density into *three* parts:

$$\mathbf{J} = \mathbf{J}_f + \mathbf{J}_b + \mathbf{J}_p = \mathbf{J}_f + \boldsymbol{\nabla} \times \mathbf{M} + \frac{\partial \mathbf{P}}{\partial t}. \tag{7.60}$$

Gauss's law can now be written as

$$\boldsymbol{\nabla} \cdot \mathbf{E} = \frac{1}{\epsilon_0}(\rho_f - \boldsymbol{\nabla} \cdot \mathbf{P}),$$

or

$$\boldsymbol{\nabla} \cdot \mathbf{D} = \rho_f, \tag{7.61}$$

where, as in the static case,

$$\mathbf{D} \equiv \epsilon_0 \mathbf{E} + \mathbf{P}. \tag{7.62}$$

Meanwhile, Ampère's law (with Maxwell's term) becomes

$$\boldsymbol{\nabla} \times \mathbf{B} = \mu_0 \Big(\mathbf{J}_f + \boldsymbol{\nabla} \times \mathbf{M} + \frac{\partial \mathbf{P}}{\partial t} \Big) + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t},$$

or

$$\boldsymbol{\nabla} \times \mathbf{H} = \mathbf{J}_f + \frac{\partial \mathbf{D}}{\partial t}, \tag{7.63}$$

where, as before,

$$\mathbf{H} \equiv \frac{1}{\mu_0}\mathbf{B} - \mathbf{M}. \tag{7.64}$$

Faraday's law and $\boldsymbol{\nabla} \cdot \mathbf{B} = 0$ are not affected by our separation of charge and current into free and bound parts, since they do not involve ρ or $\mathbf{J}$.

In terms of *free* charges and currents, then, Maxwell's equations read

$$\boxed{\begin{array}{ll} \text{(i)} \quad \boldsymbol{\nabla} \cdot \mathbf{D} = \rho_f, & \text{(iii)} \quad \boldsymbol{\nabla} \times \mathbf{E} = -\dfrac{\partial \mathbf{B}}{\partial t}, \\ \text{(ii)} \quad \boldsymbol{\nabla} \cdot \mathbf{B} = 0, & \text{(iv)} \quad \boldsymbol{\nabla} \times \mathbf{H} = \mathbf{J}_f + \dfrac{\partial \mathbf{D}}{\partial t}. \end{array}} \tag{7.65}$$

Some people regard these as the "true" Maxwell's equations, but please understand that they are in *no way* more "general" than Eq. 7.49; they simply reflect a convenient division of charge and current into free and nonfree parts. And they have the disadvantage of hybrid notation, since they contain both $\mathbf{E}$ and $\mathbf{D}$, both $\mathbf{B}$ and $\mathbf{H}$. They must be supplemented, therefore, by appropriate **constitutive relations**, giving $\mathbf{D}$ and $\mathbf{H}$ in terms of $\mathbf{E}$ and $\mathbf{B}$. These depend on the nature of the material; for linear media,

$$\mathbf{P} = \epsilon_0 \chi_e \mathbf{E} \quad \text{and} \quad \mathbf{M} = \chi_m \mathbf{H}, \tag{7.66}$$

so

$$\mathbf{D} = \epsilon\mathbf{E} \quad \text{and} \quad \mathbf{H} = \frac{1}{\mu}\mathbf{B}, \tag{7.67}$$

where $\epsilon \equiv \epsilon_0(1 + \chi_e)$ and $\mu \equiv \mu_0(1 + \chi_m)$. You may remember that **D** is called the electric "displacement"; that's why the second term in the Ampère/Maxwell equation (iv) came to be called the **displacement current**. In this context,

$$\mathbf{J}_d \equiv \frac{\partial \mathbf{D}}{\partial t}. \tag{7.68}$$

Problem 7.42. Sea water at frequency $\nu = 4 \times 10^8$ Hz has permittivity $\epsilon = 81\epsilon_0$, permeability $\mu = \mu_0$, and resistivity $\rho = 0.23\,\Omega \cdot \mathrm{m}$. What is the (amplitude) ratio of conduction current to displacement current in sea water, at this frequency? [*Hint:* Consider a parallel-plate capacitor immersed in sea water and driven by a voltage $V_0 \cos(2\pi\nu t)$.]

7.3.6 Boundary Conditions

In general, the fields **E**, **B**, **D**, and **H** will be discontinuous at a boundary between two different media, or at a surface that carries a charge density σ or a current density **K**. The explicit form of these discontinuities can be deduced from Maxwell's equations (Eq. 7.65), in their integral form

$$\left.\begin{aligned} &\text{(i)} \quad \oint_{\mathcal{S}} \mathbf{D} \cdot d\mathbf{a} = Q_{f_{\text{enc}}}, \\ &\text{(ii)} \quad \oint_{\mathcal{S}} \mathbf{B} \cdot d\mathbf{a} = 0, \end{aligned}\right\} \quad \text{over any closed surface } \mathcal{S}.$$

$$\left.\begin{aligned} &\text{(iii)} \quad \oint_{\mathcal{P}} \mathbf{E} \cdot d\mathbf{l} = -\frac{d}{dt}\int_{\mathcal{S}} \mathbf{B} \cdot d\mathbf{a}, \\ &\text{(iv)} \quad \oint_{\mathcal{P}} \mathbf{H} \cdot d\mathbf{l} = I_{\text{enc}} + \frac{d}{dt}\int_{\mathcal{S}} \mathbf{D} \cdot d\mathbf{a}, \end{aligned}\right\} \quad \text{for any surface } \mathcal{S} \text{ bounded by the closed loop } \mathcal{P}.$$

Applying (i) to a tiny, wafer-thin Gaussian pillbox extending just slightly into the material on either side of the boundary (Fig. 7.49), we obtain

$$\mathbf{D}_{\text{above}} \cdot \mathbf{a} - \mathbf{D}_{\text{below}} \cdot \mathbf{a} = \sigma_f a.$$

(The positive direction for **a** is "upward.") The edge of the wafer contributes nothing in the limit as the thickness goes to zero; nor does any *volume* charge density.) Thus, the component of **D** that is perpendicular to the interface is discontinuous in the amount

$$\boxed{D^{\perp}_{\text{above}} - D^{\perp}_{\text{below}} = \sigma_f.} \tag{7.69}$$

Identical reasoning, applied to equation (ii), yields

$$\boxed{B^{\perp}_{\text{above}} - B^{\perp}_{\text{below}} = 0.} \tag{7.70}$$

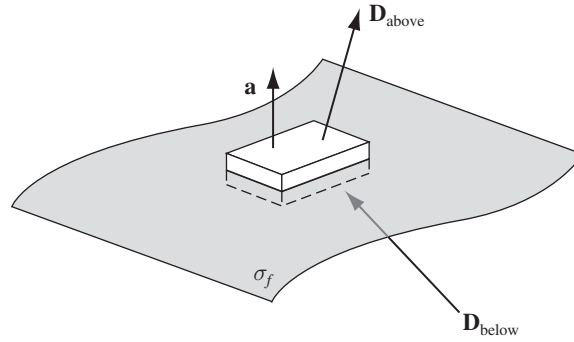


Fig. 7.49

Turning to (iii), a very thin Ampèrian loop straddling the surface gives

$$\mathbf{E}_{\text{above}} \cdot \mathbf{l} - \mathbf{E}_{\text{below}} \cdot \mathbf{l} = -\frac{d}{dt} \int_{\mathcal{S}} \mathbf{B} \cdot d\mathbf{a}.$$

But in the limit as the width of the loop goes to zero, the flux vanishes. (I have already dropped the contribution of the two ends to $\oint \mathbf{E} \cdot d\mathbf{l}$, on the same grounds.) Therefore,

$$\boxed{\mathbf{E}^{\parallel}_{\text{above}} - \mathbf{E}^{\parallel}_{\text{below}} = \mathbf{0}.} \tag{7.71}$$

That is, the components of $\mathbf{E}$ *parallel* to the interface are continuous across the boundary. By the same token, (iv) implies

$$\mathbf{H}_{\text{above}} \cdot \mathbf{l} - \mathbf{H}_{\text{below}} \cdot \mathbf{l} = I_{f_{\text{enc}}},$$

where $I_{f_{\text{enc}}}$ is the free current passing through the Ampèrian loop. No *volume* current density will contribute (in the limit of infinitesimal width), but a *surface* current can. In fact, if $\hat{\mathbf{n}}$ is a unit vector perpendicular to the interface (pointing "upward"), so that $(\hat{\mathbf{n}} \times \mathbf{l})$ is normal to the Ampèrian loop (Fig. 7.50), then

$$I_{f_{\text{enc}}} = \mathbf{K}_f \cdot (\hat{\mathbf{n}} \times \mathbf{l}) = (\mathbf{K}_f \times \hat{\mathbf{n}}) \cdot \mathbf{l},$$

and hence

$$\boxed{\mathbf{H}^{\parallel}_{\text{above}} - \mathbf{H}^{\parallel}_{\text{below}} = \mathbf{K}_f \times \hat{\mathbf{n}}.} \tag{7.72}$$

So the *parallel* components of $\mathbf{H}$ are discontinuous by an amount proportional to the free surface current density.

Equations 7.69–7.72 are the general boundary conditions for electrodynamics. In the case of *linear* media, they can be expressed in terms of $\mathbf{E}$ and $\mathbf{B}$ alone:

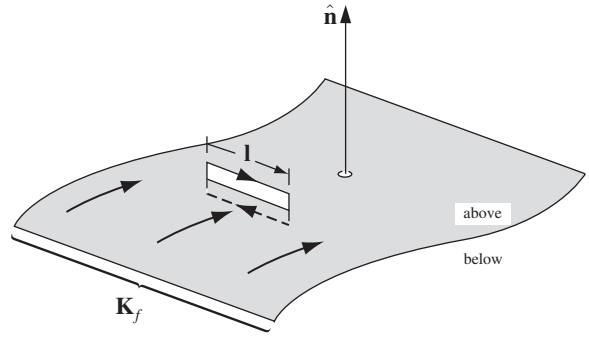


Fig. 7.50

$$\begin{aligned}
&\text{(i)} \quad \epsilon_{\text{above}} E^{\perp}_{\text{above}} - \epsilon_{\text{below}} E^{\perp}_{\text{below}} = \sigma_f, \\
&\text{(ii)} \quad B^{\perp}_{\text{above}} - B^{\perp}_{\text{below}} = 0, \\
&\text{(iii)} \quad \mathbf{E}^{\parallel}_{\text{above}} - \mathbf{E}^{\parallel}_{\text{below}} = \mathbf{0}, \\
&\text{(iv)} \quad \frac{1}{\mu_{\text{above}}} \mathbf{B}^{\parallel}_{\text{above}} - \frac{1}{\mu_{\text{below}}} \mathbf{B}^{\parallel}_{\text{below}} = \mathbf{K}_f \times \hat{\mathbf{n}}.
\end{aligned} \tag{7.73}$$

In particular, if there is no free charge or free current at the interface, then

$$\begin{aligned}
&\text{(i)} \quad \epsilon_{\text{above}} E^{\perp}_{\text{above}} - \epsilon_{\text{below}} E^{\perp}_{\text{below}} = 0, \\
&\text{(ii)} \quad B^{\perp}_{\text{above}} - B^{\perp}_{\text{below}} = 0, \\
&\text{(iii)} \quad \mathbf{E}^{\parallel}_{\text{above}} - \mathbf{E}^{\parallel}_{\text{below}} = \mathbf{0}, \\
&\text{(iv)} \quad \frac{1}{\mu_{\text{above}}} \mathbf{B}^{\parallel}_{\text{above}} - \frac{1}{\mu_{\text{below}}} \mathbf{B}^{\parallel}_{\text{below}} = \mathbf{0}.
\end{aligned} \tag{7.74}$$

As we shall see in Chapter 9, these equations are the basis for the theory of reflection and refraction.

7.4 The Field of a Rotating Magnet

Imagine an axially symmetric magnet rotating at constant angular velocity about its axis – a cylindrical bar magnet, if you like, or for that matter the Earth itself (pretending that the magnetic axis coincides with the rotation axis). It is natural to wonder whether its magnetic field is stationary, or rotates with the magnet (or, as Arlo Guthrie would say, "Maybe it doesn't even matter"). Are the field lines "dragged along" by the rotation of the source? Faraday did a series of experiments (variations on the disk experiment of Ex. 7.4) that appeared to settle the issue. Mounting the magnet along the axis of the disk he first held the magnet still while he rotated the disk (that's the original version), and a current was generated in the circuit (Fig. 7.51). Then he allowed the magnet to rotate at the same rate as the disk; again a current was generated. Finally, he held the disk at rest and rotated the magnet. This time *no* current flowed. He concluded

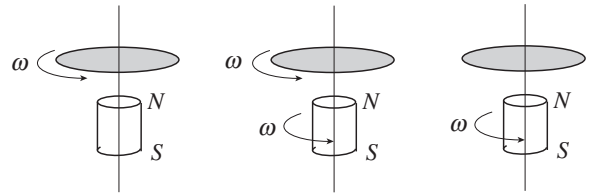


Fig. 7.51

that the field of the rotating magnet is the same as that of the stationary magnet: that's why the first two experiments give the same result, and the third is just what you'd get if the magnet were at rest. Apparently a magnet rotating about its axis of symmetry does *not* drag the field around as it spins; the field lines are stationary.[34]

And yet, if the magnet rotates about an axis *perpendicular* to its axis of symmetry, then it certainly *does* drag the field along with it. The field lines loop around from the north pole to the south pole, and that direction keeps changing. So not everyone accepted Faraday's conclusion, until 1912 when experiments by S. J. Barnett finally led to a consensus that Faraday was right – in particular, the Earth's magnetic field (or more precisely the part of it associated with rotation about the geographic north-south axis) does *not* rotate with the Earth.[35]

But what exactly does it *mean* to say that the field lines do (or do not) rotate with their source? Faraday had a very concrete conception of field lines as a network of "wires" (or cobwebs) existing "out there" in space. In that case, to say that they are moving (or not) makes perfect sense. But in Maxwell's theory it is *fields* that have real physical existence, and field *lines* are just constructs we devise (by joining up the arrows that represent the fields themselves) to help us visualize (and draw) the fields. And we have already learned that even *static* magnetic field lines can be problematic (Section 5.3.4).

Consider for a moment the electrical analog. Imagine a point charge at rest at the origin. Its (electric) field "radiates" out in all directions, and it is (obviously) constant in time (Fig. 7.52). The field lines are like spokes on a

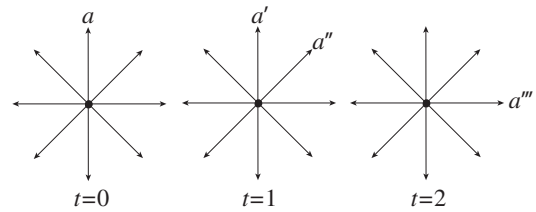


Fig. 7.52

[34] If the magnetic field lines were moving, they would presumably have induced an electric field, just as they did in Fig. 7.21b.

[35] For the full history, see C. F. Chyba and K. P. Hand, *Phys. Rev. Appl.* **6**, 014017 (2016).

wheel (or rather pins in a pin cushion, in three dimensions). A moment later the picture looks exactly the same. But suppose I ask *which* field line at $t = 1$ corresponds to the field line I labeled a at $t = 0$? It's a', obviously (I hear you say). Not so fast! Why not a''? That would still be a perfectly good field-line pattern, representing exactly the same field. The *labeling* of the field lines is completely arbitrary, with no physical content at all. If you prefer to say that the field line called a (at $t = 0$) "corresponds" to the field line a'' at $t = 1$, and then to a''' at $t = 2$, and so on, more power to you! For you, the field lines rotate, even though the source does *not*. It's not *wrong*, exactly (it describes the field correctly at every moment), it's just *dumb* – a complicated and confusing way to think about something that is intrinsically simple and straightforward.

But suppose instead of a point charge we have a uniformly charged *sphere*, and it's (uniformly) *rotating*. The electric field is exactly the same (for points outside), and it's still static (because the charge *density* is constant in time). It is now accompanied by a (static) *magnetic* field, but that doesn't alter the *electric* field. Now, however, it is not quite so dumb to wonder whether the electric field lines rotate with the sphere. Well, let 'em rotate, if it pleases you (you can even have them rotate at some completely different rate, about some other axis, if you want to be really perverse); this has nothing to do with *physics* – they represent the same electric *field* – it just has to do with how you choose to direct the field-line movie. Arlo was right: it doesn't even matter. Field lines tell you two things: their density tells you the magnitude of the field, and the arrow tells you its direction. As long as they get those two things right, you can't ask for more, because that's *all there is to know* about the fields. *Which* field line at one moment corresponds to which field line at some other moment … that's a metaphysical question; it has no "right" answer, because it is devoid of physical content. And for electric fields that's pretty much the end of the story.

But the *magnetic* case is more pernicious, because of the $\mathbf{v}$ in the Lorentz force law. That's the velocity of the charge … but relative to *what*? Nowadays we would answer "relative to whatever inertial reference frame you choose to work in – the frame in which the magnetic field is $\mathbf{B}$." But if you conceive of magnetic field lines as "wires" out there in space, moving around in time, it is easy to persuade yourself (at any rate, it was in the nineteenth century) that $\mathbf{v}$ must be the velocity of the charge *relative to the field line* at that location. (Remember those electrical engineers talking about "cutting" field lines.) And if that is the case (which, I hasten to say, it is *not*[36]) then it matters critically whether the field lines are dragged along with the rotation of the magnet. If they *are* dragged along, then Faraday should have found the same current when the magnet rotated and the disk was at rest as he did when the disk rotated and the magnet was at rest, and

[36] Footnote 34 was just plain *wrong*: moving magnetic field lines don't induce electric fields – time-dependent *fields* do.

he should have seen *no* current when the two rotated together. But that's the opposite of what he actually observed, so he concluded that the field lines are *not* dragged along by the rotation of the magnet. And similarly Barnett concluded that the Earth does not drag its magnetic field around as it rotates. It's not an unreasonable inference, and in practice it is sometimes an illuminating way to picture how the field evolves in time. That's why geophysicists, plasma physicists, and astrophysicists use this language. But it is a *choice*, not something forced by the laws of electrodynamics, so when you read about field lines being "dragged" around, ask yourself what precisely the author *means* by the statement, and remind yourself that field lines don't come in different colors (or carry serial numbers) so you could keep track of individual lines from frame to frame in the movie.

More Problems on Chapter 7

! **Problem 7.43.** Two long, straight copper pipes, each of radius a, are held a distance $2d$ apart (see Fig. 7.53). One is at potential V_0, the other at $-V_0$. The space surrounding the pipes is filled with weakly conducting material of conductivity σ. Find the current per unit length that flows from one pipe to the other. [*Hint:* Refer to Prob. 3.14.]

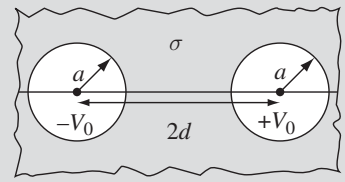


Fig. 7.53

! **Problem 7.44.** A rare case in which the electrostatic field $\mathbf{E}$ for a circuit can actually be *calculated* is the following:[37] Imagine an infinitely long cylindrical sheet, of uniform resistivity and radius a. A slot (corresponding to the battery) is maintained at $\pm V_0/2$, at $\phi = \pm\pi$, and a steady current flows over the surface, as indicated in Fig. 7.54. According to Ohm's law, then,

$$V(a,\phi) = \frac{V_0\phi}{2\pi} \quad (-\pi < \phi < +\pi).$$

(a) Use separation of variables in cylindrical coordinates to determine $V(s,\phi)$ inside and outside the cylinder. [*Answer:* $(V_0/\pi)\tan^{-1}[(s\sin\phi)/(a+s\cos\phi)]$ (for $s < a$); $(V_0/\pi)\tan^{-1}[(a\sin\phi)/(s+a\cos\phi)]$ (for $s > a$).]

(b) Find the surface charge density on the cylinder. [*Answer:* $(\epsilon_0 V_0/\pi a)\tan(\phi/2)$.]

[37] See M. A. Heald, *Am. J. Phys.* **52**, 522 (1984). See also J. A. Hernandes and A. K. T. Assis, *Phys. Rev. E* **68**, 046611 (2003).

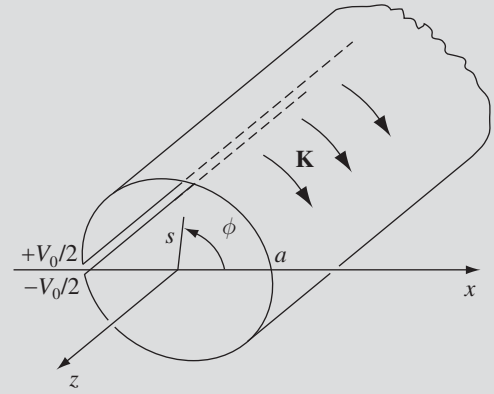


Fig. 7.54

Problem 7.45. The magnetic field outside a long straight wire carrying a steady current I is

$$\mathbf{B} = \frac{\mu_0}{2\pi}\frac{I}{s}\hat{\boldsymbol{\phi}}.$$

The *electric field inside* the wire is uniform:

$$\mathbf{E} = \frac{I\rho}{\pi a^2}\hat{\mathbf{z}},$$

where ρ is the resistivity and a is the radius (see Exs. 7.1 and 7.3). *Question:* What is the electric field *outside* the wire?[38] The answer depends on how you complete the circuit. Suppose the current returns along a perfectly conducting grounded coaxial cylinder of radius b (Fig. 7.55). In the region $a < s < b$, the potential $V(s, z)$ satisfies Laplace's equation, with the boundary conditions

$$\text{(i)}\ \ V(a, z) = -\frac{I\rho z}{\pi a^2}, \qquad \text{(ii)}\ \ V(b, z) = 0.$$

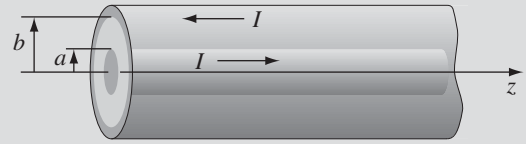


Fig. 7.55

This does not suffice to determine the answer – we still need to specify boundary conditions at the two ends (though for a *long* wire it shouldn't matter much). In the literature, it is customary to sweep this ambiguity under the rug by simply *stipulating* that $V(s, z)$ is proportional to z: $V(s, z) = zf(s)$. On this assumption:

(a) determine $f(s)$;
(b) find $\mathbf{E}(s, z)$;
(c) calculate the surface charge density $\sigma(z)$ on the wire.

[38] This is a famous problem, first analyzed by Sommerfeld and others, and known in its most recent incarnation as **Merzbacher's puzzle**. See A. Sommerfeld, *Electrodynamics*, Academic Press, New York (1952), p. 125; E. Merzbacher, *Am. J. Phys.* **48**, 178 (1980); further references in R. N. Varnay and L. H. Fisher, *Am. J. Phys.* **52**, 1097 (1984).

[*Answer:* $V = (-Iz\rho/\pi a^2)[\ln(s/b)/\ln(a/b)]$. This is a peculiar result, since E_s and $\sigma(z)$ are *not* independent of z – as one would certainly expect for a truly *infinite* wire.]

Problem 7.46. Refer to Prob. 7.21 and Fig. 7.21b. As a quantitative model,[39] let the magnetic field be uniform ($-B_0\hat{\mathbf{z}}$) in the shaded region (width w), but taper off linearly in a slab of width d at the left- and right-hand surfaces (Fig. 7.56). Choose axes with x horizontal (to the right), y vertical (upward), and z coming out of the page, with the origin at the center of the left end of the wire loop. Set the clock so the left edge of the magnet passes the origin at $t = 0$.

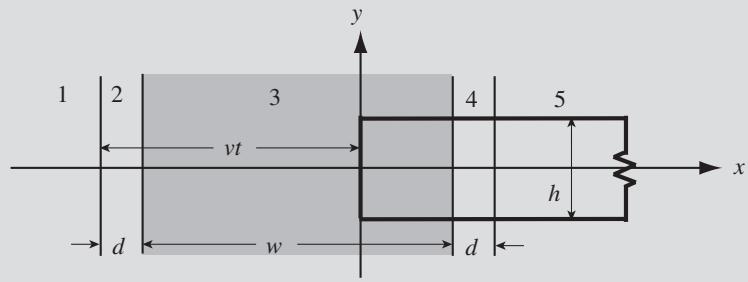


Fig. 7.56

(a) What is $\mathbf{B}(x, t)$ in each of the five regions? (Let the magnet extend indefinitely in the y and z directions.)
(b) Differentiate to get $\partial\mathbf{B}/\partial t$, and specify the regions in which it is nonzero.
(c) Find the induced electric field in region 3, and the resulting emf in the loop. Is it what you expected?

Problem 7.47. In a **perfect conductor**, the conductivity is infinite, so $\mathbf{E} = \mathbf{0}$ (Eq. 7.3), and any net charge resides on the surface (just as it does for an *im*perfect conductor, in electro*statics*).

(a) Show that the magnetic field is constant ($\partial\mathbf{B}/\partial t = \mathbf{0}$), inside a perfect conductor.
(b) Show that the magnetic flux through a (stationary) perfectly conducting loop is constant.

A **superconductor** is a perfect conductor with the additional property that (the constant) $\mathbf{B}$ inside is in fact *zero*. (This "flux exclusion" is known as the **Meissner effect**.[40])

(c) Show that the current in a superconductor is confined to the surface.
(d) Superconductivity is lost above a certain critical temperature (T_c), which varies from one material to another. Suppose you had a sphere (radius a) above its critical temperature, and you held it in a uniform magnetic field $B_0\hat{\mathbf{z}}$ while cooling it below T_c. Find the induced surface current density $\mathbf{K}$, as a function of the polar angle θ.

[39] This problem was suggested by Martin Ligare.
[40] The Meissner effect is sometimes referred to as "perfect diamagnetism," in the sense that the field inside is not merely *reduced*, but canceled entirely. However, the surface currents responsible for this are *free*, not bound, so the actual *mechanism* is quite different.

Problem 7.48. A familiar demonstration of superconductivity (Prob. 7.47) is the levitation of a magnet over a piece of superconducting material. This phenomenon can be analyzed using the method of images.[41] Treat the magnet as a perfect dipole $\mathbf{m}$, a height z above the origin (and constrained to point in the z direction), and pretend that the superconductor occupies the entire half-space below the xy plane. Because of the Meissner effect, $\mathbf{B} = \mathbf{0}$ for $z \leq 0$, and since $\mathbf{B}$ is divergenceless, the normal (z) component is continuous, so $B_z = 0$ just *above* the surface. This boundary condition is met by the image configuration in which an identical dipole is placed at $-z$, as a stand-in for the superconductor; the two arrangements therefore produce the same magnetic field in the region $z > 0$.

(a) Which way should the image dipole point ($+z$ or $-z$)?

(b) Find the force on the magnet due to the induced currents in the superconductor (which is to say, the force due to the image dipole). Set it equal to Mg (where M is the mass of the magnet) to determine the height h at which the magnet will "float." [*Hint:* Refer to Prob. 6.3.]

(c) The induced current on the surface of the superconductor (the xy plane) can be determined from the boundary condition on the *tangential* component of $\mathbf{B}$ (Eq. 5.76): $\mathbf{B} = \mu_0(\mathbf{K} \times \hat{\mathbf{z}})$. Using the field you get from the image configuration, show that

$$\mathbf{K} = -\frac{3mrh}{2\pi(r^2 + h^2)^{5/2}}\hat{\boldsymbol{\phi}},$$

where r is the distance from the origin.

! **Problem 7.49.** If a magnetic dipole levitating above an infinite superconducting plane (Prob. 7.48) is free to rotate, what orientation will it adopt, and how high above the surface will it float?

Problem 7.50. A perfectly conducting spherical shell of radius a rotates about the z-axis with angular velocity ω, in a uniform magnetic field $\mathbf{B} = B_0\hat{\mathbf{z}}$. Calculate the emf developed between the "north pole" and the equator. (Imagine that there are sliding contacts, as in Fig. 7.15.) [*Answer:* $\frac{1}{2}B_0\omega a^2$.]

! **Problem 7.51.** Refer to Prob. 7.11 (and use the result of Prob. 5.45): How long does it take a falling *circular* ring (radius a, mass m, resistance R) to cross the bottom of the magnetic field B, at its (changing) terminal velocity?

Problem 7.52.

(a) Referring to Prob. 5.55a and Eq. 7.27, show that

$$\mathbf{E} = -\frac{\partial \mathbf{A}}{\partial t}, \tag{7.75}$$

for Faraday-induced electric fields. Check this result by taking the divergence and curl of both sides.

[41] W. M. Saslow, *Am. J. Phys.* **59**, 16 (1991).

(b) A spherical shell of radius R carries a uniform surface charge σ. It spins about a fixed axis at an angular velocity $\omega(t)$ that changes slowly with time. Find the electric field inside and outside the sphere. [*Hint:* There are *two* contributions here: the Coulomb field due to the charge, and the Faraday field due to the changing $\mathbf{B}$. Refer to Ex. 5.11.]

Problem 7.53. Electrons undergoing cyclotron motion can be sped up by increasing the magnetic field; the accompanying electric field will impart tangential acceleration. This is the principle of the **betatron**. One would like to keep the radius of the orbit constant during the process. Show that this can be achieved by designing a magnet such that the average field over the area of the orbit is twice the field at the circumference (Fig. 7.57). Assume the electrons start from rest in zero field, and that the apparatus is symmetric about the center of the orbit. (Assume also that the electron velocity remains well below the speed of light, so that nonrelativistic mechanics applies.) [*Hint:* Differentiate Eq. 5.3 with respect to time, and use $F = ma = qE$.]

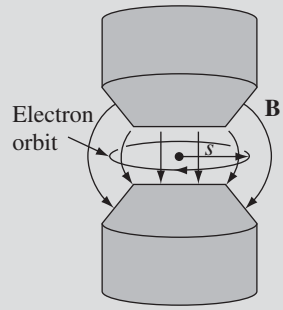


Fig. 7.57

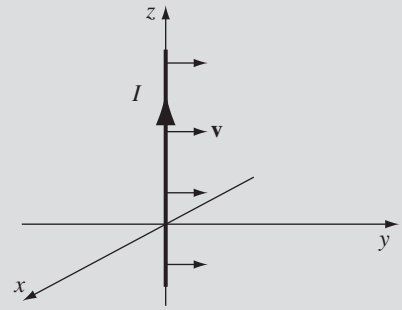


Fig. 7.58

Problem 7.54. An infinite wire carrying a constant current I in the $\hat{\mathbf{z}}$ direction is moving in the y direction at a constant speed v. Find the electric field, in the quasistatic approximation, at the instant the wire coincides with the z-axis (Fig. 7.58). [*Answer:* $-(\mu_0 Iv/2\pi s)\sin\phi\,\hat{\mathbf{z}}$.]

Problem 7.55. An atomic electron (charge q) circles about the nucleus (charge Q) in an orbit of radius r; the centripetal acceleration is provided, of course, by the Coulomb attraction of opposite charges. Now a small magnetic field dB is slowly turned on, perpendicular to the plane of the orbit. Show that the increase in kinetic energy, dT, imparted by the induced electric field, is just right to sustain circular motion *at the same radius* r. (That's why, in my discussion of diamagnetism, I assumed the radius is fixed. See Section 6.1.3 and the references cited there.)

Problem 7.56. The current in a long solenoid is increasing linearly with time, so the flux is proportional to t: $\Phi = \alpha t$. Two voltmeters are connected to diametrically opposite points (A and B), together with resistors (R_1 and R_2), as shown in Fig. 7.59. What is the reading on each voltmeter? Assume that these are *ideal* voltmeters that draw negligible current (they have huge internal resistance), and that a voltmeter registers $-\int_a^b \mathbf{E}\cdot d\mathbf{l}$ between the terminals

and through the meter. [*Answer:* $V_1 = \alpha R_1/(R_1 + R_2)$; $V_2 = -\alpha R_2/(R_1 + R_2)$. Notice that $V_1 \neq V_2$, even though they are connected to the same points![42]]

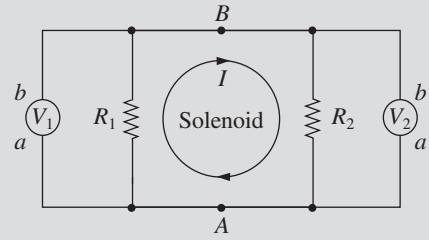


Fig. 7.59

Problem 7.57. A circular wire loop (radius r, resistance R) encloses a region of uniform magnetic field, B, perpendicular to its plane. The field (occupying the shaded region in Fig. 7.60) increases linearly with time ($B = \alpha t$). An ideal voltmeter (infinite internal resistance) is connected between points P and Q.

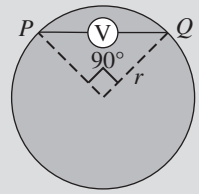


Fig. 7.60

(a) What is the current in the loop?
(b) What does the voltmeter read? [*Answer:* $\alpha r^2/2$.]

Problem 7.58. In the discussion of motional emf (Section 7.1.3) I assumed that the wire loop (Fig. 7.10) has a resistance R; the current generated is then $I = vBh/R$. But what if the wire is made out of perfectly conducting material, so that R is *zero*? In that case, the current is limited only by the back emf associated with the self-inductance of the loop (which would ordinarily be negligible in comparison with IR). As a model, let's simply replace the resistor by an inductor, L. Show that in this régime the loop (mass m) executes simple harmonic motion, and find its frequency.[43] [*Answer:* $\omega = Bh/\sqrt{mL}$.]

Problem 7.59.

(a) Use the Neumann formula (Eq. 7.32) to calculate the mutual inductance of the configuration in Fig. 7.37, assuming a is very small ($a \ll b, a \ll z$). Compare your answer to Prob. 7.23.

[42] See R. H. Romer, *Am. J. Phys.* **50**, 1089 (1982). See also H. W. Nicholson, *Am. J. Phys.* **73**, 1194 (2005); B. H. McGuyer, *Am. J. Phys.* **80**, 101 (2012).
[43] For a collection of related problems, see W. M. Saslow, *Am. J. Phys.* **55**, 986 (1987); R. H. Romer, *Eur. J. Phys.* **11**, 103 (1990).